

My Impression Of India's 2-WHEELER EV INDUSTRY

Recently, I was part of a panel discussion related to the EV industry. It was interesting to observe the EV industry terrain, especially the leadership of the 2-wheeler industry. Here are my observations and views, which are purely personal



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For starters, I was a bit surprised to see that most startup founders were from software or similar industries. Surprisingly, I didn't meet a single person with an automotive or related background. Most of them spoke about the motors, power, batteries, etc, but their overall vehicle-designs seemed to be an assembly of bought out components.

One team claimed 10-15% improvement with the components used in regular EV scooters. But they had used alloy wheels of a motor bike and the 10% improvement, I think, came from the larger circumference of the wheels. In fact, the critical controller parts were supplied by a would-be vendor but imported from, you know where!

Most of them used battery packs that were not smart, and the ones that were

supposed to have intelligence did not provide the critical data.

Most startups talked about AI and ML that could be used once the data was available in the cloud. But who was going to give that data? No one seemed to have a clue.

Charging and charging infrastructure were ignored completely. No one seemed to have any idea how much the charging impacts overall performance of the EVs.

Most startups didn't factor the cost of the battery pack, whenever it was to be replaced. If a user wanted to use the vehicle for five years, the user had to plan for one battery pack replacement, whose cost could be anywhere between 45-55% of the vehicle cost. That one element alone can be a deal-breaker when it comes to the total cost of ownership.

There was no discussion at all about hybrid vehicles. Considering the power grid situation likely in India for the next ten years, an all-electric future seems like a Mission Impossible.

I didn't see a single proposal for a vehicle that was truly hybrid. I did see one with an IC engine and BLDC motor in the front-wheel hub, but that's neither a hybrid nor will you get any benefit from it.

I have been working with a truly hybrid technology developing startup and, having been involved since the very early stages, I am aware of the complexities involved.

I hope India's EV industry will not repeat the mistakes of Indian mobile phone brands and let 'them' wipe our brands out of the market. I believe the key to success lies in the design of the controller and the software. If these are outsourced, so would be the future of your company! **EFY**

